LESSON PLAN

AIRCRAFT HUSBANDRY, CORROSION PREVENTION AND SURFACE PROTECTION (25)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Cable identification, specifications, replacement, swaged type terminals splicing, swaged terminals splicing, swaged ball terminals, NICO pr
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Cable identification, specifications, replacement, swaged type terminals splicing, swaged terminals splicing, swaged ball terminals, NICO press process and thimble eye.
5. Specific Objectives	25.1.1.1 Trainee will be able to describe the use of various standards, identification of bolts, nuts, safety methods and maintenance of cables.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of NICO press process and thimble eye installation; Safety precautions include PPE usage and tool inspection; Instructor supervision required during swaging operations.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary safety hazards associated with aircraft cable maintenance?
(b) How do you identify the correct cable specification for a given application?
(c) Why is proper terminal identification critical before beginning a splice?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Cable identification and specifications • Safety methods for cable handling • Swaged type terminals splicing • Swaged ball terminals installation	L & Ds	PPT, IB	15 min
(b)	• NICO press process overview • Thimble eye preparation and fitting • Torque settings for swaging tools • Inspection of completed splices	L & Ds	PPT, PDF	20 min
(c)	• Replacement procedures for damaged cables • Final verification of splice integrity • Documentation of maintenance actions • Common errors in terminal installation	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the correct sequence for installing a swaged ball terminal.
(b) Describe the key steps in the NICO press process for thimble eyes.
(c) What safety checks must be performed before releasing a swaged cable?

12. Home Assignment and Follow Up	Prepare a brief report on the differences between standard swaged terminals and NICO press terminals, referencing the next lesson plan on Bowden cable construction.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Cable system inspection, cable maintenance, fittings, pulleys and wear patterns, Construction and servicing of Bowden cable and Teleflex controls	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min